



YOUR KINDLE NOTES FOR:

The Wandering Earth

by Cixin Liu

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155 Highlights | 2 Notes

Highlight (Yellow) | Location 82

For instance, North China Mountain 794 – where we are now – exerts fifteen billion tons of thrust upon the earth when operating at full capacity.’

Highlight (Yellow) | Location 94

Even before the Braking Era ended, its effects upon the Earth had become horrifyingly apparent. The tides caused by the acceleration of the Earth Engines engulfed two-thirds of the Northern hemisphere’s major cities. Then, the rise in global temperatures melted the polar ice caps, which turned the flooding into a catastrophe that spread to the Southern hemisphere. Thirty years ago, my grandfather witnessed giant hundred-meter waves inundating Shanghai.

Highlight (Yellow) | Location 130

Over three centuries ago, astrophysicists discovered that the conversion rate of hydrogen to helium in the interior of the Sun was accelerating. They launched thousands of probes straight into the Sun to investigate, and eventually developed a precise mathematical model of the star. Using this model, supercomputers calculated that the Sun had already evolved away from the main sequence on the Hertzsprung-Russell diagram. Helium would soon permeate the Sun’s core, triggering a violent explosion called a helium flash. Afterward, the Sun would become a massive, cool-burning red giant, swelling until its diameter encompassed the Earth’s orbit. But our planet would have been vaporized in the preceding helium flash long before then. All of this was projected to occur in the next four hundred years. Since then, three hundred and eighty years had passed. This solar catastrophe would not only raze and consume every inhabitable terrestrial planet in the solar system – it would also completely transform the composition and orbits of the Jovian planets. After the first helium flash, as heavy elements re-accumulated in the Sun’s core, further runaway nuclear explosions would occur repeatedly for a period of time. While this period represented only a brief phase of stellar evolution, it might last thousands of times longer than all of human history. As long as we remained in the solar system, humanity stood no chance of surviving such a catastrophe. Interstellar emigration was our only way out. Given the level of technology available to humanity at the time, the only viable target for this migration was Proxima Centauri. It was the star closest to our own, a mere 4.3 light-years away.

Highlight (Yellow) | Location 154

Global support for the Takers had been rising of late, and they had grown unruly again. ‘Why do you hate us so much?’ asked Ms Stella. Several Leaver children immediately shouted in protest. ‘We won’t wait to die on Earth with you Taker fools!’ ‘We will build spaceships and depart! All hail spaceships!’

Highlight (Yellow) | Location 184

Children, only an ecosystem the size of Earth, with its unstoppable ecological cycle, could sustain us indefinitely! If humanity leaves Earth behind,’ she proclaimed, ‘then we would be as vulnerable as an infant separated from its mother in the middle of a desert!’

Highlight (Yellow) | Location 189

Humanity’s escape was a five-step process. First, the Earth Engines would generate thrust in the opposite direction of the Earth’s movement, halting its rotation. Second, operating at full capacity, the engines would accelerate the Earth until it reached escape velocity, flinging it from the solar system. Third, the Earth would continue to accelerate as it flew through outer space toward Proxima Centauri. Fourth, the engines would reverse direction, restarting the Earth’s rotation and decelerating gradually. Fifth, the Earth would enter into orbit around Proxima Centauri, becoming its satellite. People called these five steps the ‘Braking Era’, the ‘Deserting Era’, the ‘First Wandering Era’ (during acceleration), the ‘Second Wandering Era’ (during deceleration), and the ‘Neosolar Era’. The entire migration process was projected to last 2,500 years, over one hundred generations.

Highlight (Yellow) | Location 205

On our voyage home, I saw the first signal for Earth’s departure. A giant comet appeared in the night sky – the Moon. Because we could not take the Moon with us, engines had been installed on the lunar surface to push it out of Earth’s orbit, ensuring that there would be no collision during the acceleration period. The sweeping tail of the Lunar Engines bathed the sea in blue light, obscuring the stars. As it moved past, the Moon’s gravitational pull raised towering breakers. We had to transfer to a plane to fly home to the Northern hemisphere. The day of departure had finally arrived! As soon as we disembarked, we were blinded by the glare of the Earth Engines. They blazed many times brighter than before, no longer slanted but pointing straight toward the sky. The engines were running at maximum power. The planet’s acceleration created thunderous, hundred-meter waves that battered every continent. Blistering hurricanes howled through the towering columns of plasma, whipping up boiling froth and uprooting whole forests ... Our planet had become a gigantic comet, its blue tail piercing the darkness of space. Earth was on its way; humanity was on its way.

Highlight (Yellow) | Location 242

We began our new lives underground. Located five hundred meters below the surface, our city had space for over one million residents. Many others just like it were scattered across every continent. Here, I finished primary school and entered secondary school. My schooling concentrated on science and engineering. Art, philosophy, and other subjects deemed inessential had been minimized or removed from the curriculum. Humanity had no time for distractions. It was the busiest era in human history. Everyone had work to do, and the

work was never finished. Interestingly, every world religion had vanished without a trace overnight. People finally realized that if God truly existed, he was a real bastard. We still studied history, but to us, the Ante-solar Era of human history seemed as mythical as the Garden of Eden.

Highlight (Yellow) | Location 248

My father served in the Air Force as an astronaut. He frequently flew low-Earth orbit missions and was rarely at home. I remember in the fifth year of orbital acceleration, when Earth was at aphelion, we took a family trip to the seashore. Aphelion Day was a holiday like New Year's Eve or Christmas. As Earth entered the part of its orbit furthest from the Sun, everyone basked in a false sense of security, though. We still needed to wear special thermal suits to go to surface. Instead of cooling suits, we donned sealed heating suits powered by nuclear batteries. Outside, we were nearly blinded by the Earth Engines' towering plasma beams. The harsh light eclipsed our view of the surface world, and it was difficult to tell if the landscape had changed at all. We had to fly for a long time in our car before we escaped the glare and we could actually see the shore. The Sun had shrunk to the size of a baseball.

Highlight (Yellow) | Location 256

A large crowd of revelers had gathered atop the frozen sea. Fireworks whistled through the darkness. Everyone was carousing with unusual abandon. Drunken party-goers rolled across the ice, while others belted out the words to a dozen different songs, each trying to drown out the competing voices around them. 'Despite it all, everyone is living their own lives. No harm in that,' my father said approvingly. He paused, suddenly remembering something. 'Oh, I forgot to tell you – I've fallen in love with Stella Li. I want to move out to be with her.' 'Who is she?' my mother asked calmly. 'My primary school teacher,' I answered for him. I had started secondary school two years ago, and had no idea how my father knew Ms Stella. Maybe they had met at my graduation ceremony? 'Then go,' said my mother. 'I'm sure I'll grow tired of her soon enough. I'll come back then. Is that okay by you?' 'If you want to, certainly.' Her voice was as calm and even as the frozen sea. But a moment later, she bubbled with excitement. 'Oh, that one is beautiful! It must have a holographic diffractor inside!' She pointed to a firework blossoming in the night sky, genuinely moved by its beauty.

Highlight (Yellow) | Location 273

Two months later, my father returned from his jaunt with Ms Stella. My mother was neither happy nor unhappy to see him. 'Stella has a good impression of you,' my father told me. 'She said you were a very creative student.' 'Who said that?' My mother asked with a puzzled expression. 'My primary school teacher, Ms Stella,' I replied impatiently. 'Dad was living with her for the last two months!'

Highlight (Yellow) and Note | Location 331

Earth swept past perihelion and climbed toward aphelion for the sixth time.

closest point in elliptical orbit to the sun orbited.

Highlight (Yellow) | Location 348

Beginning in the sixth orbital period onward, catastrophic magma seepage events occurred all too frequently in cities around the world.

Highlight (Yellow) | Location 412

‘Earth is about to pass through the asteroid belt, and the Western hemisphere is facing right toward it.’ ‘But we passed through the asteroid belt on the last few orbits. It’s no big deal, is it?’ ‘We just swiped the edges of the asteroid belt. The Space Fleet could handle that, of course. They have lasers and nukes to clear small rocks from the Earth’s path. But this time...’ He paused. ‘Haven’t you seen the news? This time, Earth will pass straight through the middle of the belt! The fleet will deal with the small rocks, but the large ones...’ On the flight back to Asia, Kayoko turned to me and asked, ‘Are those asteroids very big?’ My father was one of the Space Fleet officers tasked with asteroid diversion and destruction. Therefore, though the government had imposed the usual media blackout to prevent mass panic, I still had some idea of what was about to happen. I told Kayoko that some of the asteroids we faced were the size of mountains; even fifty-megaton thermonuclear bombs would only pockmark their surfaces. ‘They’ll have to use the most powerful weapon in the human arsenal,’ I added mysteriously. ‘You mean antimatter bombs?’ she asked. ‘What else could it be?’ ‘What is the fleet’s cruising range?’ ‘Currently their strength is limited. My dad told me it extends out to about one and a half million kilometers,’ I answered. Kayoko gave a little squeal. ‘Then we’ll be able to see it!’ ‘Best not to look.’

Highlight (Yellow) | Location 502

Three years later, as the dust in the atmosphere dissipated, humanity made its last pass through perihelion. As we reached it, those living in the Eastern hemisphere were privileged to witness the fastest sunrise and sunset in Earth’s history. The Sun leapt up from the sea and streaked rapidly across the sky. Shadows changed directions so quickly that they looked like second hands sweeping across the faces of countless clocks. It was the shortest day Earth had ever seen, over in less than an hour. When the Sun plunged below the horizon and darkness fell across the planet, I felt a twinge of grief. This fleeting day seemed like a brief summary of Earth’s four-and-a-half-billion-year history in the solar system. Even until the end of the universe, Earth would never return. ‘It’s dark,’ Kayoko said sadly. ‘The longest night,’ I replied. In the Eastern hemisphere, this night would last twenty-five hundred years. One hundred generations would pass before the light of Proxima Centauri illuminated this continent anew. The Western hemisphere was facing its longest day, but even so, it would last just a moment compared to our age-long night. On that side of the world, the Sun would quickly rise to its zenith, where it would remain motionless, steadily shrinking. Within half a century, it would be difficult to distinguish from any other star.

Highlight (Yellow) | Location 527

I knew that Jupiter was thirteen hundred times the size of Earth, but only when I saw its immense splendor did I truly take in its colossal size.

Highlight (Yellow) | Location 532

The famous Great Red Spot appeared, still raging across Jupiter's surface after hundreds of thousands of years. The maelstrom was large enough to swallow three Earths. Jupiter now filled the entire sky. Earth was like a balloon floating on Jupiter's boiling red sea of clouds. The Great Red Spot climbed to the middle of the sky and stared down upon our world like a cyclopean eye. The entire landscape was shrouded in its ghastly light. It was impossible to believe that our tiny planet could escape the gravitational field of this colossus.

Highlight (Yellow) | Location 545

As it departed Jupiter, Earth reached escape velocity. It no longer needed to return toward the Sun, where only death lurked. As it hurtled toward the open reaches of space, the endless Wandering Era began.

Highlight (Yellow) | Location 550

After we left Jupiter behind, Asia's ten thousand Earth Engines roared to life again. They would operate at full capacity for the next five hundred years, constantly accelerating the planet. During those five hundred years, the engines would consume half of the mountains on the Asian continent as fuel.

Highlight (Yellow) | Location 577

An NGO even launched an array of probes toward the Sun, which passed through their target three months later. The data transmitted by the probes finally confirmed the fact: The Sun had not changed at all in the past four centuries.

Highlight (Yellow) | Location 663

The helium flash had occurred.

Highlight (Yellow) | Location 669

The solar explosion lasted only a short time. After two hours, the light rapidly weakened until it was extinguished altogether. A dim red sphere had replaced the Sun. From our vantage point, it slowly swelled until it reached the size of the Sun of old, a strange memory from Earth's original orbit. It was so voluminous that its diameter exceeded the orbit of Mars. Mercury, Venus, Mars – Earth's constant companions – had been reduced to wisps of smoke by the intense thermal radiation. But it was no longer our Sun. No longer emitting light and heat, it resembled a cold piece of red paper pasted onto the firmament, its muted glow merely a reflection of the surrounding starlight. This was the evolutionary fate common to all midsized stars: transformation into a red giant. Five billion years of majestic life were now a fleeting dream. The Sun had died.

Highlight (Yellow) | Location 680

As I recall all of this now, half a century has passed. Twenty years ago, Earth sailed past Pluto's orbit and out of the solar system, continuing its lonely voyage into the vast, cold reaches of space. My last visit to the surface

was a dozen or so years ago. I was accompanied by my son and my daughter-in-law, a blonde-haired, blue-eyed girl. She was pregnant at the time. When we arrived on the surface, the first thing I noticed was that I could no longer see the Earth Engines' massive plasma beams, even though I knew the engines were still operating at full capacity. The Earth's atmosphere had vanished, leaving nothing to scatter the plasma's light. The ground was covered with strange translucent yellow-green crystals. They were made of solid oxygen and nitrogen, the remnants of our frozen atmosphere.

Highlight (Yellow) and Note | Location 690

0.5 percent of light speed.

~3M mph

Highlight (Yellow) | Location 1015

We raised a mountain. You came and climbed it. 'Mountain climbing is my passion,' Fan answered. >> It is not a question of passion; we must climb mountains.

Highlight (Yellow) | Location 1033

We believed the rock stretched endlessly in all directions. Therefore, we saw our world as a hollow bubble in this solid universe, and so we gave our world the name Bubble World and we call this cosmology the Solid Universe Theory.

Highlight (Yellow) | Location 1040

'So these bubbles were the planets of your solid universe? Very interesting,' Fan commented. 'Density in our universe is entirely the inverse. Your universe must be an almost exact negative of the real universe.' >> The real universe? You are ignorantly considering the universe only as you know it right now. You have no idea what the real universe is like, and neither do we. Chastened, Fan decided to continue his line of enquiry. 'Was there light, air and water in your world?' >> No, none; and we needed none of them. Our world was made entirely of solids. There were no gases or liquids. 'No gases or liquids. How did you survive?' Fan asked. >> We are a mechanical life form. Our muscles and bones are made of metals, our brains are like highly integrated chips, and electricity and magnetism are our blood. We ate the radioactive rocks of our world's core, and they provided us with the energy we needed to survive. We were not created; we evolved naturally from extremely simple, single-celled mechanical life forms when – by pure chance – the radioactive energies formed p–n junctions in the rocks. Instead of your use of fire, our earliest ancestors discovered the use of electromagnetism. In fact, we never found fire in our world. 'It must have been very dark there then,' Fan remarked. >> Actually, there was some light. It was generated by the radioactive activity within our world's walls. Those walls were our sky. The light of that 'sky' was very weak, and it constantly shifted as the radioactivity fluctuated. Yet it led us to evolve eyes.

Highlight (Yellow) | Location 1056

The evolutionary process of mechanical life is much slower than that of carbon-based life, but eventually we reached the same ends by different means; and so one day we, too, came to contemplate our universe.

Highlight (Yellow) | Location 1061

Exploration for us meant tunneling into the walls in an attempt to find other bubbles in our solid universe. We had spun many fascinatingly alluring myths around these distant spaces, and almost all of our literature dealt with the fantasy of other bubbles. Soon, however, exploration was outlawed – forbidden on pain of death by short-circuiting. ‘Outlawed? By your church?’ Fan assumed. >> No, we have no church. A civilization that cannot see the sun and stars will be without religion. There was a very practical reason for our senate to forbid tunneling: we were not furnished with the near-infinite space you have at your disposal. Our existence was limited to that 3,500-mile bubble. All the debris that the tunneling produced ended up within this space.

Highlight (Yellow) | Location 1104

To their great surprise, the data indicated that the rock density had steadily decreased the further the ship had dug. ‘Your world had no gravity. How did you ever measure density?’ Fan interjected. >> We used inertia; it’s somewhat more complicated than your methods.

Highlight (Yellow) | Location 1111

If the density of the Bubble World continually decreased as one dug outward, then it stood to reason that it would eventually reach zero. Using the gathered data, our scientists were easily able to calculate that this would happen at about twenty thousand miles. ‘Oh, that sounds very much like how Hubble used the redshift!’ Fan exclaimed, recognizing the concept. >> It is indeed very similar. Since you could not conceive of the redshift velocity exceeding the speed of light, you concluded that it denoted the edge of the universe; and it was very easy for our ancestors to comprehend that an area with a density of zero is open space.

Highlight (Yellow) | Location 1116

it was assumed that density decreased in proportion to distance from the Bubble World, eventually declining to the point of opening into a space that would continue into infinity. This is known as the Open Universe Theory. >> The Solid Universe Theory, however, was deeply ingrained in our culture, and its supporters dominated the discourse.

Highlight (Yellow) | Location 1256

In the eighth year after the end of the war, the Helix became the first ship to cross the remaining 2,250 miles, completing the five-thousand-mile journey from the planet’s center, 3,250 miles from the edge of the Bubble World. ‘Wow! That was all the way to the surface! It must have been so exciting for you to see the great plains and real mountains!’ Fan exclaimed, fully absorbed in the visitors’ story. >> There was nothing to be excited about; the Helix reached the seabed. Fan stared up at the alien sphere in shocked silence. >> When it happened,

the images from the seismic communicator began to shake and suddenly stopped appearing altogether. Communication had been lost. A bubble ship tunneling through the rock beneath it could only catch one strange sound on its seismoscopes; a noise that in the open would have sounded like something being peeled. It was the sound of tons upon tons of water bursting into the vacuum of the Helix. Neither the machine life forms nor the technology of the Bubble World had ever been designed to come into contact with water. The powerful electric current produced by short-circuiting life and equipment almost instantly vaporized everything

Highlight (Yellow) | Location 1318

The People of the Core had accepted this as the amorphous rock's natural state. Following this discovery, the ship continued to descend, and as it did the temperature on board began to rise. >> It was with a deep fear that the explorers suddenly came face to face with the most horrible realization yet: the amorphous rock was alive! Stirring, its surface had begun to roil with anger, its terrifying form now covered with countless bursting bubbles. But this monster's surging life force seemed to consume its very being, its body dissolving into a ghostly white shadow. Once all the amorphous rock in the container had transformed into this new phantasmal state, the explorers began to feel a strange sensation grip their bodies. Within moments, the sparks of shorting circuits erupted from within their bodies, ending their lives in agonizing fireworks. >> Seismic waves transmitted this terrible spectacle live to the Explorer Alliance, right up until the monitors, too, went dark. A quickly dispatched relief ship suffered the same fate. As soon as it made contact with the doomed vessels, its crew also erupted into terrible sparks, dying in pain. It seemed as if the amorphous rock had become a specter of death, looming over all of space. The scientists, however, noticed that the second series of short circuits was nowhere as violent as the first explosive deaths. This led them to a conclusion: as the area of space increased, the density of that amorphous shadow of death decreased. >> It took many lives and countless horrible deaths, but in the end the People of the Core finally discovered another state of being they had never encountered before: gas.

Highlight (Yellow) | Location 1336

Even though we came to an ever greater understanding of water vapor, we still lacked the sealing technology that would have allowed the core's scientists to protect our people and machinery from harm.

Highlight (Yellow) | Location 1339

To study the strange new states, the World Government and the Explorer Alliance constructed a laboratory at an altitude of 2,900 miles. They equipped this facility with a permanent drainpipe. Here experts began to study the amorphous rock in earnest. 'Only then could you begin to undertake the work of Archimedes,' Fan chimed in. >> You are quite correct, but you should not forget that our earliest forebears had already done the work of Faraday. >> As a byproduct of their work in the Laboratory for Amorphous Rock Research, our scientists came to discover water pressure and buoyancy. They also managed to develop and perfect the sealant technology necessary to deal with liquids.

Highlight (Yellow) | Location 1346

All that would be required was a sufficiently sealed and pressure-resistant vessel. Without excavators, this ship would be able to rise at speeds that seemed almost incomprehensible to the People of the Core. 'You built a

Bubble World rocket,' Fan noted with a smile. >> More of a torpedo, really. This torpedo was a metallic, pressure-resistant, egg-shaped container with no drive or propeller whatsoever.

Highlight (Yellow) | Location 1373

Gagarin carefully opened the torpedo's hatch and slowly raised himself out of the vessel. He immediately felt the gust of the ocean breeze, and, after a few perplexed moments, he realized that it was gas. Tremors of fear shook his body as he recalled a flow of water vapor he had once seen through a steel-rock pipe in the laboratory. Who could have ever foreseen that there could be this much gas anywhere in the universe? Gagarin soon understood that this gas was very different from the gas produced by boiling amorphous rock. Unlike the latter, it did not cause his body to short circuit. >> In his memoirs he later wrote the following description of these events: >> I felt the gentle touch of a giant invisible hand brush by my body. It seemed to have reached down from a vast, boundless, and completely unknown place; and that place was now before me, transforming me into something wholly new. >> Gagarin lifted his head, and then and there he finally embraced the reward of one hundred thousand years of our civilization's exploration: he saw the magnificent, sparkling wonder of the starlit sky.

Highlight (Yellow) | Location 1386

'It really wasn't easy for you. You had to explore for so many years, just to reach our starting point!' Fan exclaimed in admiration. >> That is the reason why you should consider yourselves a very lucky civilization.

Highlight (Yellow) | Location 1394

'Why did you come here?' he asked the sphere above. >> We are just passing by, and we wanted to see if there was intelligent life here with which we could have a chat. We talk to whoever first climbs this mountain. 'Where there's a mountain, there will always be someone to climb it,' Fan intoned, nodding. >> Indeed, it is the nature of intelligent life to climb mountains, to strive to stand on ever higher ground to gaze farther into the distance. It is a drive completely divorced from the demands of survival. Had you, for example, only been concerned with staying alive, you would have fled from this mountain as fast and as far away as you could. Instead, you chose to come and climb it. The reason evolution bestows all intelligent life with a desire to climb higher is far more profound than mere base needs, even though we still do not understand its real purpose. Mountains are universal, and we are all standing at the foot of mountains. 'I am on the top of the mountain,' Feng Fan interjected. He would not stand for anyone, not even aliens, challenging the glory of his having climbed the world's tallest mountain. >> You are standing at the foot of the mountain. We are all always at the foot. The speed of light is the foot of a mountain; the three dimensions of space are the foot of a mountain. You are imprisoned in the deep gorge of light-speed and three-dimensional space. Does it not feel ... cramped?

Highlight (Yellow) | Location 1409

Does it feel small to you? 'How could it? The universe stretches out endlessly before my eyes; scientists can even peer as far as twenty billion light years into space,' Fan explained. >> Then I shall tell you: it is no more than a bubble world twenty billion light years in radius. Fan had no words.

Highlight (Yellow) | Location 1412

Our universe is an empty bubble, a bubble within something more solid. 'How could that possibly be? Would this larger solid not immediately collapse in under its own gravity?' Fan asked, bewildered. >> No, at least not yet. Our bubble is still expanding in this super-universal solid. Gravitational collapse is only an issue for a bounded, solid space. If, however, the surrounding solid area is actually limitless, then gravitational collapse would be a non-issue. This, of course, is no more than a guess. Who could know whether this solid super-universe has its own limits?

Highlight (Yellow) | Location 1418

one could theorize that at its immense scale, gravity is offset by some other force, just like electromagnetism is largely offset by nuclear forces on the microscopic scale. We are not aware of such a force, but when we were inside the Bubble World we remained unaware of gravity.

Highlight (Yellow) | Location 1421

'What is this solid? Is it...?' Fan hesitated for a moment. 'Rock?' he finally asked. >> We do not know, but we will discover that in fifty thousand years, once we reach our destination. 'Where exactly are you going?' Fan asked. >> The edge of the universe. Our bubble ship is called the Needle's Point. Do you remember the name? 'I remember,' Fan answered. 'That was the ship that first discovered the law of decreasing density in the Bubble World.' >> Correct. We do not know what we will find. 'Does the super-universe have other bubbles in it?' Fan enquired. >> You are already thinking very far ahead, indeed. 'How could I not?' Fan responded. >> Think of the many small bubbles inside a very big rock. They are there, but they are very hard to find. Even so, we will go and look for them. 'You truly are amazing,' Fan smiled, feeling a deep admiration for the adventurous aliens.

Highlight (Yellow) | Location 1433

It was a pleasure meeting you; and remember, mountains are universal.

Highlight (Yellow) | Location 1458

What kind of mountain was left for him to climb? He would have to survive; he had to find out. The primal fear of the Himalayan blizzard returned. Once, this fear had made him cut the rope connecting him to his companions and his lover. He had sealed their fate. Now he knew that he had done the right thing. If there had been anything left for him to betray to save his life, he would have betrayed it. He had to live. There was a universe of mountains out there.

Highlight (Yellow) | Location 1476

When Pa felt miserable, he would crouch over his pipe for hours, unmoving, as if he had become a clod of dirt on the yellow earth.

Highlight (Yellow) | Location 1479

Was the face of an old farmer any different? The trees, the soil, the houses, the people – everything was blackened, yellowed, wrinkled. He could not see the eyes of this face that stretched toward the horizon,

Highlight (Yellow) | Location 1481

In youth, that gaze had been filled with longing for rain; in old age, it had grown glassy.

Highlight (Yellow) | Location 1486

Life goal #1: Drink some water that is not bitter, make some money.

Highlight (Yellow) | Location 1513

Life goal #2: Go to a city with more lights and sweeter water, make more money.

Highlight (Yellow) | Location 1565

‘Modern society is full of opportunities,’ Zhuang said, leaning toward Shui. ‘The skies are thick with golden birds. Perhaps one day you will reach out and seize one, but only if you learn to take yourself seriously.’ Shui looked around, but he did not see a single golden bird. He shook his head and said, ‘I never got an education.’ ‘That is certainly regrettable, but who knows? Maybe it will turn out to your advantage in the end. The greatness of this era lies in its unpredictability. Miracles can happen to anyone.’

Highlight (Yellow) | Location 1597

Life goal #3: Go to a bigger city, see more of the world, make even more money.

Highlight (Yellow) | Location 1599

When Shui saw the capital for the first time, one thing became clear: some things had to be seen to be understood. The power of his imagination alone was inadequate.

Highlight (Yellow) | Location 1633

That’s eighteen hundred per month. I can’t go any higher than that. The base pay for this kind of work used to be four or five hundred yuan, plus additional piece work. Now we pay a fixed monthly salary, which is not bad in comparison.’ So Shui became a high-rise window cleaner, otherwise known as a ‘spiderman’.

Highlight (Yellow) | Location 1638

Life goal #4: Become a Beijinger.

Highlight (Yellow) | Location 1671

The experience taught Shui a philosophical truth: some things only became clear when seen from above. Swallowed up in the capital, everything around him seemed hopelessly complicated. On the ground, the city was like an unending labyrinth. Up here, it was nothing more than an anthill with ten million inhabitants, and the world around it was so vast!

Highlight (Yellow) | Location 1682

He pointed to a building at random and asked how much the apartment on the top floor cost. The saleslady told him it was a three-bedroom, one-den apartment and that it cost thirty-five hundred yuan per square meter, which worked out to three hundred and eighty thousand yuan in total. Shui drew a sharp gasp when he heard the number, but the woman's next statement softened the brutal figure considerably: 'You can pay by monthly installments of fifteen hundred to two thousand yuan.' 'I-I'm not from Beijing. Could I still buy it?' he asked carefully. The saleslady flashed him a winning smile. 'You are too funny. The household registration system was dismantled years ago. Is there such a thing as a "real" Beijinger anymore? If you settle down here, doesn't that make you a Beijinger?'

Highlight (Yellow) | Location 1701

The China Sun Project had begun the selection process for the critical technology at the heart of the project: the material that would be used to build its reflector. In the end, Zhuang's nano mirror film was chosen from among a dozen other materials. Overnight, the scientifically inclined vagrant was transformed into one of the chief scientists of the China Sun Project, recognized the world over.

Highlight (Yellow) | Location 1708

'To tell the truth, that morning when I told you about the opportunities of these times, I had lost faith in just about everything. I was mostly saying those things for my own benefit, but these days the world truly is brimming with opportunities!' Shui nodded too. 'There are golden birds everywhere.'

Highlight (Yellow) | Location 1719

'This is the China Sun? Shui asked in awe, pointing to the mirror. Zhuang nodded. 'It is a thirty-thousand-square-kilometer reflector. From geosynchronous orbit – at an altitude of thirty-six thousand kilometers – it will reflect sunlight onto Earth. Viewed from the surface, it will look like there is another sun in the sky.'

Highlight (Yellow) | Location 1723

How does an extra sun in the sky bring more rain?' 'This artificial sun can employ many different methods to influence the weather. For instance, by disturbing the thermodynamic equilibrium of the atmosphere, it can influence atmospheric circulation, increase ocean evaporation or shift weather fronts,' Zhang answered. 'But that doesn't really explain it. In fact, the orbital reflector is just one part of the China Sun Project. The other part is a complex model of atmospheric motion, which will run on multiple supercomputers. It will be able to

accurately simulate motion in any given region of the atmosphere and then identify a critical point. If heat from the artificial sun is brought to bear upon this point, the effect would be dramatic enough to completely transform the climate of a targeted area for a period of time.’ He paused. ‘The process is extremely complicated, and it is outside my area of expertise. I don’t quite understand it myself.’

Highlight (Yellow) | Location 1731

‘How can something so big hang in the sky without falling down?’ Zhuang gazed silently at Shui for several long seconds. Finally, he glanced down at his watch and then clapped Shui on the shoulder. ‘Let’s go. I’m treating you to dinner. I’ll explain why the China Sun will not fall while we eat.’ The explanation did not turn out to be as easy as Zhuang expected. He was forced to set aside his original topic and start with the basics. While Shui knew he lived on a round planet, the traditional Chinese model of a heavenly dome over a square Earth was still rooted deep in his mind. It required a great deal of effort from Zhuang to make Shui really understand that the world in which he lived was a small spherical rock floating through an endless void. Although Shui came no closer to understanding why the China Sun would not fall, the universe was greatly changed in his mind’s eye that evening. He entered his own Ptolemaic Era.

Highlight (Yellow) | Location 1738

The second evening, Zhuang ate dinner with Shui at a roadside food stall and successfully dragged him into the Copernican Era. Over the next two evenings, Shui slogged through the Newtonian Era, acquiring an elementary understanding of universal gravitation. The evening after that, with the help of the globe in his office, Zhuang ushered Shui into the Space Age. On the next public holiday, in front of that globe, Shui finally grasped the meaning of a geosynchronous orbit. At last, he understood why the China Sun would not fall down.

Highlight (Yellow) | Location 1779

Life goal #5: Fly to space, clean the China Sun.

Highlight (Yellow) | Location 1810

‘Every one of you has to clean a pane of glass before I let you up!’ It was an impossible demand. The people below could only cling to their safety harnesses or the ropes supporting the platforms for dear life, not daring to move. They were utterly incapable of loosening one hand to pick up a squeegee or remove the lid from the detergent bucket. Every day, these aerospace officials dealt with altitudes as high as tens of thousands of kilometers in the form of blueprints and documents; but now, as they gained a first-hand feel for four hundred meters, they were scared witless. Zhuang rose and walked to the spot above an air force colonel. Of the dozen people hanging off the side of the building, he was the only one who remained calm and collected. The colonel began to clean the glass, keeping his motions steady and controlled. What astonished Shui most, however, was that the man was working with both hands, and he had relinquished his grip on anything he might use to steady himself. Even so, his board remained motionless against the wall in the strong wind, a feat that only veteran spidersmen could accomplish. When Shui recognized the man, the scene in front of him no longer seemed as strange; he was an astronaut who had flown on the Shenzhou 8 spacecraft more than a decade earlier. ‘Colonel Zhang, in your candid opinion, is the task before you really easier than a spacewalk in orbit?’ asked Zhuang. ‘With respect to the physical ability and skill required, the difference is not great,’ the former astronaut replied.

Highlight (Yellow) | Location 1823

cleaning skyscrapers and cleaning the reflector in space. Both tasks require workers to constantly maintain their balance in the face of danger, while performing repetitive, monotonous, physically demanding labor. Both tasks require constant vigilance, as the slightest carelessness can lead to an accident. For an astronaut, that might mean deviation from orbit, lost tools or materials, or a malfunction in his life support system. For a spiderman, that might mean shattered glass, dropped tools or detergent, or breakage or slippage of his safety harness. In terms of physical strength, technical skill, and psychological fortitude, the spidermen

Highlight (Yellow) | Location 1828

The former astronaut lifted his head and nodded at Zhuang. 'I am reminded of that old parable about the oil peddler who could pour oil into a bottle through the square hole in a copper coin. He was every bit as skilled as a general who never missed a bull's-eye. The only difference between them was their social status.'

Highlight (Yellow) | Location 1848

Zhuang turned to look at the young spidermen and then told the manager in a raised voice, 'But, sir, the salary I am offering is eight to ten times better than what you pay them!' The next day, Shui and sixty of his fellow spidermen arrived at the National Aerospace Training Center in Shijingshan. Each and every one of them was a farm boy who had come from some remote corner of China's vast countryside to Beijing, looking for work.

Highlight (Yellow) | Location 1871

Shui unfastened his own seatbelt and, using swimming motions, floated clumsily through the air to the nearest porthole. For the first time, he saw Earth in its entirety with his own eyes.

Highlight (Yellow) | Location 1881

Soon, the pole came into view. All of the longitudinal seams met at a small black dot. As the shuttle began its descent toward the dot, Shui realized with a start that it was actually a gigantic tower rising above the silver plain. He knew that this hermetically sealed cylinder was the China Sun Control Station. For the next three months, it would be their only home in the desolation of space. And so the spidermen began their new lives in space. Every day – the China Sun orbited Earth once every twenty-four hours – they piloted small tractor-like machines onto the mirrored surface to polish it. They drove their tractors to and fro across the wide expanse of the reflector, as if they were tilling the silver earth. As a result, the Western media coined a more poetic name for the spidermen. They were now 'mirror farmers'.

Highlight (Yellow) | Location 1896

When they returned to the control station at the end of each day, they were often too exhausted to even take off their spacesuits. As more personnel arrived from Earth, the control station began to feel cramped, and they lived like crewmen aboard a submarine. Nonetheless, they considered themselves fortunate if they could return to the station at all. The most remote point on the reflector was nearly one hundred kilometers from the station.

Cleaners working on the reflector's outer rim often could not make it back after a day on the job and had to spend the 'night' in the 'wilderness'. After suctioning a liquid dinner from their suit, they would fall asleep suspended in space. The work was incredibly dangerous, to boot.

Highlight (Yellow) | Location 1904

On his fifth day in space, Shui received a call from his family. He was working more than fifty kilometers away from the control station, and the China Sun had its beam trained on his home village. He heard Pa's voice. 'Wah, are you on that sun? It's shining above our heads right now. The night is as bright as day!' Shui replied, 'Yeah, Pa, I'm right above you.' Then Ma spoke. 'Wah, is it hot up there?' 'You could say it's both hot and cold. Right now, everything outside of my shadow is hotter than ten summers in our village, but inside my shadow is colder than ten winters.' 'I can see our Wah,' Ma told Pa. 'That little black dot on the Sun right there!' Shui knew this was impossible, but as tears rolled down his cheeks, he said, 'Pa, Ma, I can see you, too. There are two little black dots on the Asian continent where you are! Dress warmly tomorrow. I can see a cold front moving in from the north!'

Highlight (Yellow) | Location 1922

The mirror farmers did an outstanding job. Over time, they began to take on responsibilities beyond the scope of their cleaning work. At first, they simply repaired damage done to the reflector by meteor strikes, but later they were tasked with more demanding work: monitoring and reinforcing sections at risk for overstress failure.

Highlight (Yellow) | Location 1939

it was clear to the mirror farmers that they would be the first and last group of workers in space to receive only a primary school education. Those who followed them would be university graduates at the very least. Still, they served the purpose envisioned by Zhuang. They had proven that skill, experience, and the ability to adapt to adverse circumstances were more important than knowledge and creativity in the blue-collar jobs created by space development. Ordinary people were fully up to the task. However, space did alter the way the mirror farmers thought. No one else had the privilege of gazing down upon Earth from thirty-six thousand kilometers every day. With a glance, they could take in the whole planet. To them, the global village was no longer just a metaphor but a reality before their very eyes.

Highlight (Yellow) | Location 1949

Shui bought a house in Beijing, married, and had a child. He spent half of every year at home, and the other half in space. He loved his job. His long patrols on that silver land more than thirty thousand kilometers above Earth filled his heart with detached peacefulness. He felt as though he had found his ideal life, and the future stretched before him as level and smooth as the silver plain underfoot.

Highlight (Yellow) | Location 1952

then something happened that shattered his tranquility and thoroughly changed the course of his mental journey. Shui encountered Stephen Hawking. No one had expected that Hawking would live to be one hundred. It was a medical miracle, but it was also a testament to his force of will.

Highlight (Yellow) | Location 1964

What Hawking enjoyed most about his life on the China Sun were his daily excursions across the surface of the reflector. To the consternation of many, he would simply drift along the underside of the reflector for several hours every day. Shui, who by now was the China Sun's most experienced spacewalker, was assigned to accompany the professor on his outings. At that time, Hawking's fame rivaled Einstein's – even Shui had heard of him. Nevertheless, Shui was shocked when they met for the first time in the control center. He had never imagined that someone with such a severe disability could achieve so much – not that he understood the great scientist's achievements in the slightest. On their excursions, however, Hawking betrayed no hint of his paralysis.

Highlight (Yellow) | Location 1971

Hawking found it difficult to communicate with Shui. He did have an implant that allowed him to control a speech synthesizer with his brain waves, which made speaking less of a chore than it had been in the previous century. However, his words still had to be run through a device that provided real-time translation into Chinese so that Shui could understand him. Shui's superiors instructed him that he was never to initiate conversation with the professor in case he disturbed his thoughts. Hawking, however, was more than willing to talk to him. He first asked Shui for an account of his life and then began to reminisce about his own early years.

Highlight (Yellow) | Location 1980

After a while, their conversations inevitably turned to science. Shui feared this would bring their discussions to an end, but it was not the case. For the professor, it was relaxing to discuss deep topics in physics and cosmology using language that even ordinary people could follow. He told Shui about the Big Bang, black holes, and quantum gravity. When Shui returned to the station, he began to wrestle with the thin little book Hawking had written in the previous century, consulting the station's engineers and scientists when he encountered something he did not understand. He grasped far more of its contents than anyone thought he would.

Highlight (Yellow) | Location 1988

'The world below is complicated,' agreed Shui, 'but seen from so far away, the universe seems so simple, just stars scattered in space.' 'Yes, my boy, it does indeed,' said the professor.

Highlight (Yellow) | Location 1994

By the light of the stars, the black latitude and longitude lines formed by the support beams passed beneath their feet, as if they were skimming above the surface of a tiny, tranquil planet. Whenever the reflector was reoriented or reshaped, the engines on the back face would ignite. Illuminated by countless jets of flame, the surface of this tiny planet seemed even more beautiful and mysterious. And shining above, always, the Milky Way, bright and unwavering. It was here that Shui first encountered the deepest secrets of the cosmos. He learned that the starry sky that filled his vision was but a speck of dust in the unimaginable vastness of the universe, and that this entire creation was nothing but the embers of a ten-billion-year-old explosion.

Highlight (Yellow) | Location 2000

Many years ago, when he had taken his first step as a spiderman onto the roof of a skyscraper, Shui had seen all of Beijing. When he arrived on the China Sun, he had seen all of Earth. Now, Shui faced the third such glorious moment of his life. Standing on the roof of the cosmos, he could see things beyond his wildest dreams. Although he possessed only a superficial understanding of those distant worlds, they still held an irresistible attraction for him.

Highlight (Yellow) | Location 2003

Once Shui expressed his confusion to an engineer in the station. ‘Humanity landed on the Moon in the sixties. And what next? Even now, we have not set foot on Mars. We don’t even visit the Moon anymore.’ ‘Humans are practical creatures,’ replied the engineer. ‘What was driven by idealism and faith in the middle of the last century was not viable in the long term.’ ‘What’s wrong with idealism and faith?’ ‘There’s nothing wrong with them per se, but economic interests are better.’

Highlight (Yellow) | Location 2009

Ordinary people like you and me would never have made it to space at all, even if we are no further than near-Earth orbit. Pal, don’t let Hawking poison you. Normal folk shouldn’t toy with the things he does.’ The conversation changed Shui. On the surface he appeared calm, working as hard as ever, but deep down he was contemplating new horizons. Twenty years flew by. From an altitude of thirty-six thousand kilometers, Shui and his compatriots commanded a clear view of two decades of changes creeping across the globe.

Highlight (Yellow) | Location 2013

They watched as the Three-North Shelter Belt formed a verdant ribbon that traversed northwestern China, slowly turning the yellow desert green. Their home villages would never lack rain or snow again. The dry riverbeds on the outskirts of the villages flowed once more with clear, clean water. The China Sun deserved the credit for all of this. It had played a major role in the great campaign to transform the climate of northwestern China. Not only that, it also performed a number of innovative extracurricular activities: once it melted the snows of Mount Kilimanjaro to ease a drought in Africa; on another occasion it turned an Olympic host city into a city that truly never slept.

Highlight (Yellow) | Location 2020

The Ministry of Space Industry held a grand ceremony to decorate the first group of industrial workers in orbit. They were honored not only for their twenty years of exceptional hard work, but more importantly, these sixty men were recognized for the singular accomplishment of entering space as youths with nothing but an elementary or middle-school education. In doing so, they had thrown the doors of space development wide open to everyone. Economists unanimously agreed that this had been the true beginning of the industrialization of space.

Highlight (Yellow) | Location 2111

‘One final question,’ Zhuang said. ‘In one thousand years’ time, as you race past stars at speeds of tens or hundreds of thousands of kilometers per second, it will take decades or even centuries for humanity to receive the weak radio signals you send out. Is it worth the sacrifice?’ With a smile, Shui announced to the whole world, ‘As the China Sun flies beyond the solar system, humans will look away from all our creature comforts and up toward the starry sky again.’

Highlight (Yellow) | Location 2117

Life goal #6: Fly to the stars, draw humanity’s gaze back to the depths of the cosmos.

Highlight (Yellow) | Location 2124

The China Sun had accomplished its goal before it ever began its journey. Humanity’s enthusiasm for exploring beyond the solar system was reborn. Zhuang’s thoughts returned to that sultry summer night in that northwestern city twenty-three years ago, when he and a farm boy from the arid countryside had boarded the night train to Beijing.

Highlight (Yellow) | Location 2128

Finally, the spot of light came to rest on northwestern China. At its center lay the little village in which Shui had been born. By the side of the road on the outskirts of town, Shui’s parents stood together with their neighbors, watching the China Sun fly east. Pa shouted into the phone, ‘Wah, you are going somewhere far away?’ ‘Yeah, Pa,’ Shui replied from space. ‘I am afraid I will not come home.’ ‘Is it very far away?’ Ma asked. ‘Very far, Ma,’ Shui answered. ‘Farther than the Moon?’ Pa asked. Shui fell silent for a few seconds. Then, in a voice much lower than before, he said, ‘Yeah, Pa, a little farther than the Moon.’

Highlight (Yellow) | Location 2145

The morning Sun also shone down on the path that led out of the village. White poplars now lined the path, and a short distance away a small river ran parallel to it. On that day twenty-four years ago, in the small hours of the morning, under the same light of dawn, the son of northwestern peasants gradually disappeared into the distance on this very road, nursing vague hopes. It was now broad daylight in Beijing, but Zhuang remained standing on the roof of the Aerospace Tower, gazing at the point where the China Sun had vanished.

Highlight (Yellow) | Location 2153

After two years, it would fly beyond Pluto’s orbit and become an interstellar spaceship, and its crew would enter hibernation. After forty years, it would fly past Alpha Centauri, and its crew would briefly reawaken. One century after the China Sun began its journey, Earth would receive information obtained during their exploration of Alpha Centauri. By then, the China Sun would already be soaring toward Sirius. Thanks to the speed boost from Alpha Centauri’s three suns, it would have reached fifteen percent of the speed of light. Another sixty years later, one hundred and sixty years after setting out from Earth, it would reach Sirius. After passing the

binary star system formed by Sirius A and Sirius B, its speed would increase to twenty percent of the speed of light, and it would hurtle even deeper into the night sky. Given the limits of the onboard cryogenic hibernation system's lifespan, the China Sun might reach ε Eridani or

Highlight (Yellow) | Location 2170

their relationship with clients should be like that of the forehead to the back of the skull; the two should never meet. This, of course, was in the best interest of both parties.

Highlight (Yellow) | Location 2180

His instructor had also said that clients were like mistresses: you could have more than one, but you should never let them meet.

Highlight (Yellow) | Location 2182

Three years had passed since the Creator civilization had left Earth. Their grand cosmic visit had drastically increased humanity's ability to mentally cope with alien civilizations. Moreover, the Creators' fleet of 20,000 spaceships had blotted out the sky, but only one ship from the Elder Brothers' world had arrived on Earth. It was not as bizarrely shaped as the spaceships of the Creators. Cylindrical with rounded ends, it looked like an intergalactic cold-relief capsule.

Highlight (Yellow) | Location 2187

When Smoothbore recognized some of the faces around the room, the magnificent hall suddenly felt shabby. The most conspicuous among them was SinoSys Group's Zhu Hanyang, whose 'Orient-3000' operating system was replacing the outdated Windows OS worldwide.

Highlight (Yellow) | Location 2195

now, these thirteen financial princelings had assembled to hire a professional hitman to kill three people, and according to his contact this was only the first batch. Smoothbore paid the diamond ring no attention. His eyes were fixed on the three photographs in Zhu's hand – clearly the units that required processing.

Highlight (Yellow) | Location 2228

The ship's smooth sides bloomed with slowly shifting iridescent patterns, producing a hypnotic effect on those who looked too long. In fact, the surface of the ship was a perfect mirror, and the patterns seen by observers on the ground were only the distorted reflections of Earth below.

Highlight (Yellow) | Location 2231

Before their departure three years ago, the Creators told humanity that they had created six Earths in total; the four that now remained were within two hundred light-years of each other. They urged the people of Earth to

devote their full efforts to technological development – we needed to eliminate our brother planets, lest we be destroyed ourselves. But this warning came too late. Locking their ship into orbit around Earth, emissaries from one of those three planets, the first Earth, arrived in the solar system not long after the departure of the Creators. The First Earth civilization was twice as old as mankind, and so the people of this Earth came to call them ‘Elder Brothers’.

Highlight (Yellow) | Location 2320

‘Do you like my paintings?’ Smoothbore glanced around and saw that most of the paintings were just chaotic splotches of color – paint splashed directly onto a canvas would have seemed rational by comparison. But there were a few pictures in a very realistic style, and Smoothbore’s eyes were quickly drawn to one of these: a canvas dominated by a cracked yellow earth. A few dead plants protruded from the fissures in the ground, looking as if they had withered away centuries ago, if indeed water had ever existed in this world at all. A skull lay on the parched earth. Though it was bleach-white and permeated with cracks, two green, living plants sprouted from its mouth and one eye-socket. In sharp contrast with the drought and death surrounding them, these plants were green and luxuriant, and a tiny, delicate flower crowned the tip of one sprout. The skull’s other eye-socket contained a human eyeball. Its limpid pupil stared at the sky, and its gaze was filled with the same wonder and loss as that of its painter. ‘I like this one,’ Smoothbore said, pointing to the painting. ‘It’s called Barren No. 2. Will you buy it?’ ‘How much?’ ‘How much you got?’ Smoothbore pulled out his wallet and removed all the hundred-yuan notes it contained. He handed them to the artist, but the latter only took two bills. ‘It’s only worth this much. It’s yours now.’

Highlight (Yellow) | Location 2350

Mingled in the rotten food, it was often possible to make out a roast suckling pig with only the legs eaten away, a barely-touched grouper, whole chickens ... Recently, it had become common to find whole Silkie hens, owing to the popularity of a new dish called White Jade Chicken. The dish was prepared by slitting open the stomach of the chicken, filling it with tofu, and letting it simmer. The slices of tofu were the real delicacy; the chicken, while delicious, was merely casing. Like the reed leaves around rice dumplings, any diner foolish enough to eat the chicken itself would become the laughingstock of more discerning epicureans

Highlight (Yellow) | Location 2384

His tutor’s words unfurled in his mind, laying out their profession’s maxim: the gun does not care at whom it is aimed.

Highlight (Yellow) | Location 2406

the most common terms in his new profession: their fundamental business was called ‘processing’, their targets were ‘units’ or ‘work’, and death was ‘cooling’.

Highlight (Yellow) | Location 2407

The academy was divided into L, M and S disciplines, or long-, mid- and short-range.

Highlight (Yellow) | Location 2409

Smoothbore's instructors advised him and his classmates to keep their distance from L students: 'They are the nobility of this profession, as they are the most likely to change the course of history.'

Highlight (Yellow) | Location 2427

We must solemnly and perfectly communicate his or her final message to the unit, and thus inflict the maximum psychic shock and torment possible prior to cooling. This is the romance and beauty of S discipline – the look of terror and despair in the unit's eyes just before cooling. We can find no greater pleasure than this in our work; but to this end, we must cultivate our verbal dexterity and literary acumen.' So, for one year, Smoothbore studied literature. He read Homer's epics, memorized Shakespeare and studied works by many other classical and contemporary authors. Smoothbore felt this was the most rewarding year of his overseas education.

Highlight (Yellow) | Location 2453

On the first day of their real training, the principal made each student present his or her weapon. As he placed Snubnose in that line of finely-crafted pistols, Smoothbore had felt deeply embarrassed. The principal, however, picked up Snubnose, hefted it in his hand, and said with sincere admiration: 'This is a fine gun.' 'It doesn't have rifling, and you can't even attach a silencer,' sneered another student. 'Precision and range are of little importance to S specialists, and rifling even less so. And silencers? A small pillow will do the trick. Boy, do not allow yourself to be limited by stale convention. In the hands of a master, this gun can yield artistry that all your expensive toys cannot.' The principal was right.

Highlight (Yellow) | Location 2461

'From now on, we will call you Smoothbore!' said the headmaster, handing the gun back to its owner. 'Hold on to this, boy. It looks like you will have to study knife-throwing.'

Highlight (Yellow) | Location 2830

we soon discovered that things were far more complicated than we had imagined, and the balance we had achieved would soon be destroyed.' 'Destroyed by what?' 'Education. You know that in the present age of Fourth Earth, education is the sole means of social ascendancy. If society is an ocean, stratified by differences in temperature and salinity, then education is a pipe that connects the ocean floor to the surface and prevents the complete isolation of each layer.' 'So you're saying that fewer and fewer poor people could afford to attend university?' 'Yes. The cost of higher education grew increasingly expensive, until it became a privilege reserved for the sons and daughters of the social elite. However, the price of traditional education did have limits, even if they were only crude market considerations, so while the pipe grew gossamer-thin, it did not vanish completely. But one day, the appearance of a dramatic new technology fundamentally changed education.' Smoothbore hazarded a guess. 'Do you mean the ability to transmit knowledge directly to the brain?' 'Yes, but the direct infusion of knowledge was only part of it. A human brain could be implanted within a supercomputer with a capacity that far exceeded that of the brain itself; the inventoried knowledge of the computer could then be recalled by the implantee as distinct memories.' The alien continued, 'But this was only one of the computer's

secondary functions. It was an amplifier of intelligence, an amplifier of understanding, and it could raise human thought to a whole new level. Suddenly, knowledge, intelligence, depth of thought – even perfection of mind, character, and aesthetic judgment – were commodities that could be purchased.’

Highlight (Yellow) | Location 2849

The pipeline from the bottom of society to the top was completely severed. Those who received this premium education were vastly more intelligent than those who did not. The cognitive differences between these educated elites and ordinary humans were as large as those between humans and dogs, and these differences manifested themselves in every aspect of human life – even artistic sensibility, for example. This super-intelligentsia formed a new culture – a culture as incomprehensible to the rest of humanity as a symphony is incomprehensible to a dog.

Highlight (Yellow) | Location 2861

‘Though the willful killing of a dog might be a punishable offense, it is not like killing a person. When human health and safety is threatened by rabies, it is judged acceptable to put all dogs down. Sympathy for poor people hinged on one shared characteristic – personhood. When the poor ceased to be people, and all commonalities between rich and poor vanished, sympathy followed suit. This was humanity’s second evolution.

Highlight (Yellow) | Location 2868

They were not governed by the Three Laws proposed by your Asimov, but by the fundamental principle of the First Earth Constitution: that private property shall be inviolable.

Highlight (Yellow) | Location 2959

In the vastness of the cosmos, is it really necessary to crowd this little cottage of mine? You have no cause to resent me. I obtained my home through perfectly reasonable and legitimate means. I got my start as the manager of a small feminine hygiene products company, and to this day, I have relied only on my own business savvy. I am a law-abiding citizen, so the Social Machine has protected and will continue to protect me and my legal property. It will not tolerate your wrongdoing, however, so I advise everyone to get going as soon as possible. Out of respect for our common evolutionary origin, I will remember you, and I hope you will remember me. Take care.

Highlight (Yellow) | Location 3030

On the day the Curse was born, science fiction author Cixin Liu visited Taiyuan on business for the 264th time. Although it was the city he hated most in the world, he always paid a visit to a small shop in the red-light district to buy a bottle of lighter fluid for his archaic Zippo lighter. It was one of the very few things he could not buy on Taobao or eBay. Snow had fallen two days prior, and, like always, it was quickly packed down into a blackened crust of ice. Cixin slipped and fell painfully on his backside. When he arrived at the train station, the pain in his ass caused him to forget to move the little bottle of lighter fluid from his travelling bag to his pocket. As a result, it was discovered during the security check, and after it was confiscated he was fined 200 yuan. He loathed this city.

Highlight (Yellow) | Location 3061

After Vista, Microsoft was hard-pressed to justify frequent upgrades to its operating system, which prolonged the life of Curse 1.0 for a time. But operating systems were like the wives of new-made billionaires: upgrades were inevitable.

Highlight (Yellow) | Location 3213

Perhaps it was the officer's intervention, or perhaps it was their utter inebriation, but three parameters escaped their alterations: 'Taiyuan, Shanxi Province, China.' Thus, Curse 4.0 was born. Taiyuan had been cursed. At the instant of its creation, the new version of the Curse fully understood the grand mission with which it had been entrusted. Because of the immensity of the task before it, Curse 4.0 did not immediately spring to action. Instead, it gave itself time to penetrate and propagate. Once it was thoroughly entrenched throughout the web, it considered its plan of attack: it would start by eliminating soft targets, then transition to hard targets and escalate things from there.

Highlight (Yellow) | Location 3302

Two beautiful young women from the SFK editorial department rode atop the horses, their long hair trailing behind them. The riders pulled Cixin and Pan up onto the backs of the horses. Then, like lightning, they took off through the gaps in the blistering sea, vaulting the burning wreckage of cars. A moment later, the smoke cleared from their vision. The horses had galloped onto the bridge that spanned the Fen River. Cixin and Pan took deep breaths of the clean, cool air. Holding the slender waists of the young women and enjoying the tickle of hair against their faces, the men lamented that their flight had not been longer.

Highlight (Yellow) | Location 3309

Cixin, Pan and the SFK staff spotted a single cyclist among the ranks of the survivors. His presence was noteworthy for a single reason: in this day and age bicycles were connected to and controlled by the web, and the Curse had locked their wheels as soon as it began its assault. The cyclist was an old man, the man once known as Sa Bi. Thanks to the Curse's early campaign of harassment, Sa Bi had developed an instinctive fear and abhorrence of the web. He had minimized his exposure to it in his daily life – by riding a twenty-year-old antique bicycle, for instance.

Highlight (Yellow) | Location 3321

Founded in 497 BCE as the capital of the state of Jin, the city had survived the turbulence of the Spring and Autumn period and the Warring States period. During the Tang Dynasty, Taiyuan waxed in importance as a strategic military stronghold in Northern China. The city was razed by Song troops in 979 CE, but it rose again, flourishing throughout the Song, Jin, Yuan, Ming and Qing dynasties. It was not just a city of great military significance but also a renowned hub of culture and trade. The suggested slogan for the city's anniversary festivities was 'Celebrating 2,500 years of Taiyuan!' But now, the city that had survived twenty-five centuries had been reduced to ashes by a sea of flames.

Highlight (Yellow) | Location 3329

Back at the Shanxi Bureau of Artificial Intelligence Safety, a senior director opened his laptop computer. The screen displayed the most recently compiled version of the virus; Curse 5.0. The target parameters for ‘Taiyuan, Shanxi Province, China’ now read ‘*, *, *’. The Micro-Era

Highlight (Yellow) | Location 3358

There was a joke that became popular in those days. It was a conversation with God, and it went like this: ‘Oh, God, for you thousands of years are just a brief moment!’ God answered, ‘Indeed, they are just a second to me.’ ‘Oh, God, for you vast riches are just small change!’ God answered, ‘Just a nickel.’ ‘Oh, God, please spare me a nickel!’ To which God then answered, ‘Certainly. Just give me a second.’

Highlight (Yellow) | Location 3373

The UNS Ark sailed for twenty-three years. However, as she traveled close to light speed, this ‘Ark Time’ equated to twenty-five thousand years on Earth. Had it followed its mission plan, the UNS Ark should have returned to Earth long ago. Flying close to the speed of light made communication with Earth impossible. Only by reducing its velocity to less than half the speed of light could the Ark be contacted by Earth. This maneuver, however, cost significant amounts of time and energy, and therefore the Ark would usually only do this once a month in order to receive a dispatch from Earth.

Highlight (Yellow) | Location 3570

It was in that darkest night that an idea came to life in the mind of a certain genetic engineer: what if humanity’s size could be reduced by nine orders of magnitude?’

Highlight (Yellow) | Location 3747

She was handed a box that one of her attendants had been carrying. She opened the box, and immediately a strange animal emerged. It was a round football-sized organism with tiny, chaotically twitching legs. The High Counsellor lifted the creature by one of its small legs and explained. ‘Look, this is one of our city’s gifts to you! A lacto-chicken!’ The Forerunner strained his mind trying to recall his microbiology education. ‘Are you saying that that is a...’ He paused in disbelief. ‘A lactobacillus?’ ‘That is what it was called in the macro-era. It is a creature that gives yogurt its taste. A very useful animal indeed!’ the High Counsellor replied. ‘A very useful bacterium,’ the Forerunner corrected. ‘But I now understand that bacteria cannot harm you at all.

Highlight (Yellow) | Location 3791

optimistic precisely because of our micro-society’s tiny scale. It meant that humanity’s ability to survive in this universe had been increased many millions of times over. For example,’ she said, ‘the contents of that can you just opened could feed our entire city for two years. And the can itself could supply our city with all the metal it needed for those two years.’ ‘As a macro-human, I now have a much better understanding of the enormous advantages of the micro-era. It’s all so mythical, so very epic!’ the Forerunner exclaimed in admiration. The

High Counsellor smiled and continued. 'Evolution trends toward the small. Size does not equal greatness. Microscopic life has a much easier time coexisting with nature in harmony. When the giant dinosaurs died out, their contemporaries, the ants, persisted.'

Highlight (Yellow) | Location 3995

Chapter 3 Ants

Highlight (Yellow) | Location 4071

'We still have a very long time to get along and very many things to talk about, but let us not speak of morals. In the universe, such considerations are meaningless.'

Highlight (Yellow) | Location 4095

the primary use for water plundered from planets – beyond providing basic life support – is fuel for this system. The Devourer's propulsion technology is based on the principle of recoil and the conservation of momentum; it is not some sort of strange, space-time bending nonsense.'

Highlight (Yellow) | Location 4156

'Don't assume that you can build a planetary engine just because you understand nuclear fusion. It's a long way from a firecracker to a rocket.'

Highlight (Yellow) | Location 4157

it is perfectly commonplace for a weaker civilization to become the livestock of a stronger civilization.'

Highlight (Yellow) | Location 4158

You will discover that being raised for food is a splendid life indeed. You will have no wants and will live happily to the end. Some civilizations have sought to become livestock, only to be turned down. That you should feel uncomfortable with the idea is entirely the fault of a most banal anthropocentrism.'

Highlight (Yellow) | Location 4412

It truly was a great interstellar war. In the lengthy annals of the Devourer's wars, ours was one of the most remarkable battles! To this day, all throughout our world, minstrels sing of the epic achievements of the Earth's soldiers,' Fangs stated. 'We

Highlight (Yellow) | Location 4487

Fangs again paused, deep in thought. 'What is civilization? Civilization is devouring, ceaselessly eating, endlessly expanding; everything else comes second.' The Captain, too, was deep in thought. Looking at Fangs,

he asked, 'Can the struggle for existence be the universe's only law of biological and cultural evolution? Can we not establish a self-sufficient, introspective civilization where all life exists in symbiosis? A civilization like that of the Eridanians?' Fangs answered without hesitation or pause. 'I am no philosopher; perhaps it can be done. The crux is, who will take the first step? If one's survival is based on the subjugation and consumption of others and if that should be the universe's iron law of life and civilization, then whoever first rejects it in favor of introspection will certainly perish.'

Highlight (Yellow) | Location 4514

he noticed more ants climbing about the soil beneath the grass. Raising himself, he turned to Fangs who was standing right by his side. 'When we leave, this grass and these ants will be the dominant species on Earth.' Fangs was at a loss for words. 'Earth's civilized life seems to be getting smaller and smaller. Dinosaurs, humans and now probably ants,'

Highlight (Yellow) | Location 4566

After the milk was warm, God took the bowl into the living room to watch TV without turning off the liquefied petroleum gas stove. When Qiusheng's wife, Yulian, returned from cleaning the cowshed and the pigsty, she could smell gas all over the house. Covering her nose with a towel, she rushed into the kitchen to turn off the stove, opened the window and turned on the fan. 'You old fool! You're going to get the whole family killed!' Yulian shouted into the living room. The family had switched to using liquefied petroleum gas for cooking only after they began supporting God. Qiusheng's father had always been opposed to it, saying that gas was not as good as honeycomb coal briquettes. Now he had even more ammunition for his argument. As was his wont, God stood with his head lowered contritely, his broom-like white beard hanging past his knees, smiling like a kid who knew he had done something wrong. 'I ... I took down the pot for heating the milk. Why didn't it turn off by itself?' 'You think you're still on your spaceship?' Qiusheng said, coming down the stairs. 'Everything here is dumb. We aren't like you, being waited on hand and foot by smart machines. We have to work hard with dumb tools. That's how we put rice in our bowls!' 'We also worked hard. Otherwise how did you come to be?' God said carefully.

Highlight (Yellow) | Location 4665

'Where are you from?' asked the delegate from of the United States. The old wanderer shook his head. 'A civilization only has a fixed location in her infancy. Planets and stars are unstable and change. Civilizations must then move. By the time she becomes a young woman, she has already moved multiple times. Then the civilizations will make this discovery: no planetary environment is as stable as a sealed spaceship. So they'll make spaceships their home, and planets will just be places where they sojourn. Thus, any civilization that has reached adulthood will be a starfaring civilization, permanently wandering through the cosmos. The spaceship is her home. Where are we from? We come from the ships.' He pointed up with a finger caked in dirt.

Highlight (Yellow) | Location 4690

'Civilization on Earth is currently in a stage of rapid development. If we're not hit by natural disasters beyond our ability to withstand, I think we will continue our development indefinitely,' said the Secretary-General.

‘Wrong. Think about it: every person experiences childhood, youth, middle age and old age, finally arriving at death. The stars are the same way. Indeed, everything in the universe goes through the same process. Even the universe itself will have to terminate one day. Why would civilization be an exception? No, a civilization will also grow old and die.’

Highlight (Yellow) | Location 4695

‘Different civilizations grow old and die in different ways, just like different people die of different diseases or of just plain old age. For the God Civilization, the first sign of her senescence was the extreme lengthening of each individual member’s life span. By then, each individual in the God Civilization could expect a life as long as four thousand Earth years. By age two thousand, their thoughts had completely ossified, losing all creativity. Because individuals like these held the reins of power, new life had a hard time emerging and growing. That was when our civilization became old.’

Highlight (Yellow) | Location 4701

‘The second sign of the civilization’s senescence was the Age of the Machine Cradle.’ ‘What?’ ‘By then our machines no longer relied on their creators. They operated independently, maintained themselves and developed on their own. The smart machines gave us everything we needed: not just material needs but also psychological ones. We didn’t need to put any effort into survival. Taken care of by machines, we lived as though we were lying in comfortable cradles. ‘Think about it: if the jungles of primitive Earth had been filled with inexhaustible supplies of fruits and tame creatures that desired to become food, how could apes evolve into humans?’

Highlight (Yellow) | Location 4721

‘One day, the ships told us that they were close to complete breakdown. The propulsion systems could no longer push the ships near the speed of light. The God Civilization could only drift along at a speed not even one-tenth the speed of light, and the ecological support systems were nearing collapse. The machines could no longer keep two billion of us alive. We had to find another way out.’ ‘Did you ever think that this would happen?’ ‘Of course. Two thousand years ago, the ships already warned us. That was when we began the process of seeding life on Earth so that in our old age we would have support.’ ‘Two thousand years ago?’ ‘Yes. Of course I’m talking about time on the ships. From your frame of reference, that was 3.5 billion years ago, when Earth first cooled down.’

Highlight (Yellow) | Location 4733

‘By traveling close to the speed of light, we possess almost limitless time.’

Highlight (Yellow) | Location 4742

‘There are two billion of you. That means every family on Earth would have to take in one or two of you.’ After the Secretary-General spoke, the meeting hall sank into silence. ‘Yes, yes, sorry to give you so much trouble...’

Highlight (Yellow) | Location 4788

Qiusheng's father quickly became slightly drunk on cups of vintage huangjiu. He gave God a thumbs-up. 'You're really something! To be able to create so many living things – you're truly supernatural.' God drank a lot, too, but his head was still clear. He waved his hand. 'No, not supernatural. It was just science. When biology has developed to a certain level, creating life is akin to building machines.'

Highlight (Yellow) | Location 4798

'In terms of events that are even more recent, after the disappearance of the ancient Aegean civilizations the world-creation software believed that civilization on Earth was successfully established. It ceased to perform further monitoring and micro-adjustments, like leaving a wound-up clock to run on its own. This resulted in further errors. For example, it should have allowed the civilization of ancient Greece to develop on her own and stopped the Macedonian conquest and the subsequent Roman conquest. Although both of these ended up as the inheritors of Greek civilization, the direction of Greek development was altered...'

Highlight (Yellow) | Location 4818

'To be comfortable in the cradle it's important to know a bit about philosophy, art, history and so on – just some common facts, not real learning. Many scholars on Earth right now have much deeper thoughts than our own.'

Highlight (Yellow) | Location 4909

As for Qiusheng's father, he thought God was a fraud, and in fact, this view was pretty common. Since scientists couldn't understand the Gods' scientific papers, there was no way to prove their authenticity. Maybe the Gods were playing a giant trick on the human race.

Highlight (Yellow) | Location 4922

Our scientists say that there was life on Earth even three billion years ago. Do you admit this?' God nodded. 'That estimate is basically right.' 'So you're three billion years old?' 'In terms of your frame of reference, yes. But according to the frame of reference of our ships, I'm only 3,500 years old. The ships flew close to the speed of light, and time passed much more slowly for us than for you. Of course, once in a while a few ships dropped down and decelerated enough to come to Earth so that further adjustments could be made to the evolution of life on Earth. But this didn't require much time. Those ships would then return to cruise at close to the speed of light and continue skipping over the passage of time here.' 'Bullshit,' Qiusheng's father said contemptuously. 'Dad, this is the Theory of Relativity,' Qiusheng interrupted. 'Our scientists already proved it.'

Highlight (Yellow) | Location 5021

'I'm not coming back,' Qiusheng said without looking back. 'What? Come back here! Is your head filled with shit?' Qiusheng's father followed him out of the house. 'What's the matter with you? Even if you don't want your wife and kid, how dare you leave your father?' Qiusheng stopped but still did not turn around. 'Why should I care about you?' 'How can you talk like that? I'm your father! I raised you! Your mother died early. You think

it was easy to raise you and your sister? Have you lost your mind?' Qiusheng finally turned back to look at his father. 'If you can kick the people who created our ancestors' ancestors' ancestors out of our house, then I don't think it's much of a sin for me not to support you in your old age.' He left, and Yulian and his father stood there, dumbfounded.

Highlight (Yellow) | Location 5063

'Gramps God,' Qiusheng's father said, his face sorrowful, 'your ships are all ancient. They won't last much longer. Where can you go then?' God stroked his beard and said calmly, 'It doesn't matter. Space is limitless. Dying anywhere is the same.'

Highlight (Yellow) | Location 5100

'First, you must get off this rock!' God spread out his arms toward space. His white robe danced in the fall wind like a sail. 'Where will we go?' Qiusheng's father asked in confusion. 'Begin by flying to the other planets in the solar system, then fly to other stars. Don't ask why, but use all your energy toward the goal of flying away, the farther the better. In that process, you will spend a lot of money, and many people will die, but you must get away from here. Any civilization that stays on her birth world is committing suicide! You must go into the universe and find new worlds and new homes, and spread your descendants across the galaxy like drops of spring rain.'

Highlight (Yellow) | Location 5107

'Next I will tell you a secret, a great secret.' He stared at everyone in the family with his blue eyes. His stare was like a cold wind and caused everyone's hearts to shudder. 'You have brothers.'

Highlight (Yellow) | Location 5130

This is a race of life and death to see which one among you can achieve near-light-speed space travel first. It is the only way to break through the prison of time and space.

Highlight (Yellow) | Location 5136

'When you fly into the universe at close to the speed of light to resolve the threat of your siblings, you must perform another urgent task: find a few planets suitable for life and seed them with some simple, primitive life from here, like bacteria and algae. Let them evolve on their own.'

Highlight (Yellow) | Location 5151

She left two thousand years ago at the speed of light. Who knows where she is now? Even if you repair your ship, how will you ever catch her? You told us that nothing can go faster than light.' God pointed at the sky with his cane. 'In this universe, as long as you're patient you can make any wish come true. Even though the possibility is minuscule, it still exists. I told you once that the universe was born out of a great explosion. Now gravity has gradually slowed down its expansion. Eventually the expansion will stop and turn into contraction. If

our spaceship can really fly again at close to the speed of light, then we will endlessly accelerate and endlessly approach the speed of light. This way, we will skip over endless time until we near the final moments of the universe. 'By then, the universe will have shrunk to a very small size, smaller even than Bingbing's toy ball, as small as a point. Then everything in the entire universe will come together, and she and I will also be together.
